R1.4 Entropy and spontaneity (HL)

The big picture (HL)

Higher level (HL)

Guiding question(s)

What determines the direction of chemical change?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

In our lives, we have many examples of disorder prevailing. For example, think of the last time you cleaned up and organised your room, **Figure 1**. Think of how much energy it took you to get it to that state and how easy it was for some things to be out of place again.

Figure 1. A room in two different scenarios: with and without order.

In previous subtopics, you have learned that in an exothermic reaction energy is released. In **Figure 2**, you can see the log of wood burning. It is undergoing a combustion reaction. All combustion reactions are exothermic and release energy. This energy comes from the breaking down of chemical bonds between the atoms of the substances that make up the wood.

Figure 2. As wood burns, the energy of the particles that make up the cells in wood disperses as they separate into smaller solids and gases.

Credit: anatoliy_gleb, Getty Images (https://www.gettyimages.com/detail/photo/man-and-woman-communicate-by-fire-against-royalty-free-image/1213691573)

In the example of the burning log, the substances that make up the wood, like cellulose and lignin, are in a solid state and their atoms only vibrate in place. As the wood burns, the temperature increases and the kinetic energy of the particles increases. The chemical bonds between the particles break and the elements reorganise into smaller compounds and disperse as either solids or gases.

Looking at this example you might wonder: why do the particles that were so organised in wood as a solid disperse when the wood burns? Does this happen in all reactions?

In the photo of the burning log, you can also see the night sky. Just like the particles in the wood, you might have heard that the galaxies, with all their stars and planets, are moving away from each other. The universe is expanding and as a result galaxies are dispersing. They are moving away from each other.

Is there a connection between the dispersion of particles and what happens during a chemical reaction?

Concept

Chemical reactions are spontaneous when there is an increase in the total entropy of the system and surroundings.

Prior learning

Before you study this subtopic make sure that you understand the following:

- that particles in the solid state contain less kinetic energy than particles in the liquid state which contain less than particles in the gas state (see section S1.1.2a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/)).
- the concepts of system and surroundings and the energy changes between the two (see section R1.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/energy-transfer-in-chemical-reactions-id-43469/)).
- how to calculate the enthalpy change of a reaction (see section R1.1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/enthalpy-and-heat-transfers-id-43576/)).